Academic-Period and Intersession Assessment of PM2.5 Infiltration at Various Building Level Indoor Spaces Using Distributed IAQ Monitoring and Spatial Modeling

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Abstract

Indoor fine particulate matter ($PM_{2.5}$) concentrations in academic buildings are affected by outdoor particle infiltration, heating, ventilation, and air conditioning (HVAC) operations, occupation patterns, and indoor activities. However, indoor-outdoor particle coupling can vary across spaces within the same building from differences in room size, room purpose, ventilation, occupancy, and indoor activities. This study investigates the spatial and temporal variation of indoor $PM_{2.5}$ concentrations within the ETEC building at the University at Albany by combining minute indoor air quality monitoring with multizone airflow and contaminant-transport modeling. Minute-level $PM_{2.5}$ measurements were collected at five indoor locations, including two classrooms, two conference rooms, and a café area containing a kitchen and hallway, as well as one outdoor location at the building entrance. Measurements from a regular academic period (March 22–28, 2026) were compared with measurements from an academic break period (March 14–20, 2026). A three-dimensional model of the ETEC building was developed using Rhino 8, Grasshopper, and the contam-in-ANT plug-in to simulate outdoor $PM_{2.5}$ infiltration and transport among the selected spaces. Sensor measurements showed higher average indoor $PM_{2.5}$ concentrations during the academic period than during spring break. Weekday–weekend differences were also more pronounced during spring break than during the academic period. Room-level comparisons showed substantial spatial variability, with the café and kitchen exhibiting the highest sensor-measured $PM_{2.5}$ concentrations, suggesting a contribution from indoor activities such as cooking. Simulated concentrations were consistently lower than measured concentrations, indicating that outdoor infiltration alone may not fully explain observed indoor $PM_{2.5}$ levels. Future work will incorporate more detailed building information, including room openings, interior characteristics, HVAC systems, door-opening frequency, and

building materials, as well as human-activity information derived from motion, CO_2, and light-level sensors. Ultimately, this work aims to develop a more comprehensive and adaptable IAQ modeling framework that can be applied to better understand and predict indoor air quality across a wide range of buildings for adaptive indoor climate control and thermal comfort.

1. Introduction

Indoor fine particulate matter ($PM_{2.5}$) refers to fine inhalable particles matter (PM) that are generally 2.5 micrometers and smaller. These particles are considered a significant environmental issue, as unlike PM of larger size, $PM_{2.5}$ particles retain in the human body and can penetrate deep into the respiratory system and potentially enter the bloodstream, negatively impacting human health. Exposure to particulate matter has been associated with respiratory and cardiovascular effects, including aggravated asthma, decreased lung function, heart attacks, irritation of airways and difficulty breathing. $PM_{2.5}$ can originate from both indoor and outdoor sources. Indoor sources of $PM_{2.5}$ rise from cooking activities, smoking, combustion, and human activities, while outdoor $PM_{2.5}$ originate from wildfires smoke, pollen, and vehicle emissions. In addition, indoor $PM_{2.5}$ can originate from outdoors when $PM_{2.5}$ infiltrate indoor spaces. Research modeled indoor $PM_{2.5}$ concentrations in indoor spaces, across different building types using computer models. However, indoor-outdoor particle coupling can vary across rooms with different purposes within the same building. This study characterizes $PM_{2.5}$ concentrations across varied rooms within a single academic building across different academic periods.

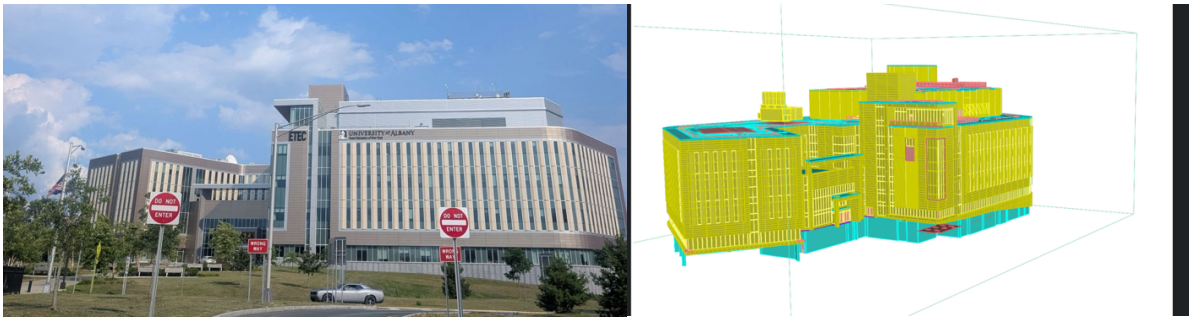


Fig 1. Photo of the ETEC Building (left) and the 3D model wireframe of the building through Rhino 8 (right)

Preliminary analysis is expected to show substantial differences in $PM_{2.5}$ levels by room type, building function, and academic period. We expect higher $PM_{2.5}$ levels in the dining/café area, where cooking and other indoor activities occur, while lower concentrations are expected in conference rooms and classrooms. Differences between weekdays and weekends are also expected to reflect changes in building occupancy and activity.



Fig 2. The view of a sensor in action (left), ETEC Basement Café (right)

Minute-level air-quality measurements were collected from five indoor locations with different functions and one outdoor location in the ETEC building at the University at Albany. The indoor locations included two small classroom/conference rooms, two large conference rooms, and a large café area containing a kitchen and hallway. The outdoor sensor was located near the building entrance. In addition to $PM_{2.5}$, the sensors measured CO_2, temperature, relative humidity, barometric pressure, motion, and light levels. These additional measurements provide information about environmental conditions and human activity that may help explain temporal and spatial variations in indoor $PM_{2.5}$. The minute-level measurements were processed and cleaned using Python and then aggregated to hourly averages for comparison between building-use periods. Two one-week periods of building use were examined: an academic break period from March 14–20, 2026, and a regular academic period from March 22–28, 2026.

To model the measured indoor $PM_{2.5}$ concentrations, a three-dimensional representation of the ETEC building was developed using Rhino 8 and Grasshopper. The contam-in-ANT

Grasshopper plug-in was used to connect the building geometry with contaminant-transport simulation framework. The building was represented as multiple airflow and contaminant zones corresponding to the monitored indoor spaces and the outdoor environment. Outdoor $PM_{2.5}$ concentrations were incorporated into the model using EPA Air Quality System data, while meteorological conditions, including temperature, atmospheric pressure, wind speed, and wind direction, were incorporated using March 2026 weather data. Transient multizone simulations were then performed to investigate the infiltration and movement of outdoor $PM_{2.5}$ through the selected indoor spaces. By comparing measured sensor concentrations with simulated concentrations, this study evaluates the contribution of outdoor infiltration while also identifying the potential influence of room characteristics, building use, and indoor activity on observed $PM_{2.5}$ patterns.

2. Methodology

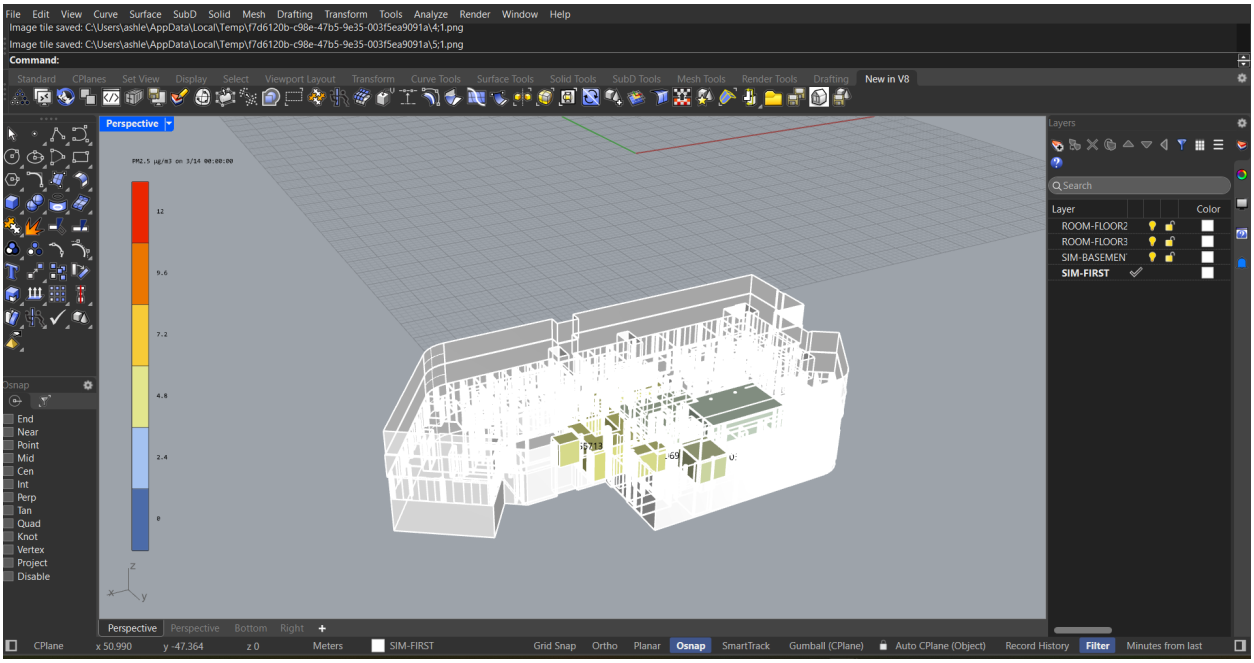


Fig 3. The interface of Rhino 8 with the ETEC building simulation model shown, simulating and visualizing $PM_{2.5}$ levels in six rooms.

Minute-level indoor air quality data were collected at five indoor locations and one outdoor location within the ETEC building at the University at Albany. Indoor measurements were obtained using Milesight AM319 Ambient Monitoring Sensors, with two sensors located in

small classrooms/conference rooms, two in large conference rooms, and one in a large café area that also includes a kitchen and hallway. Outdoor location was monitored using a Milesight AM308 Ambient Monitoring Sensor at the building entrance.

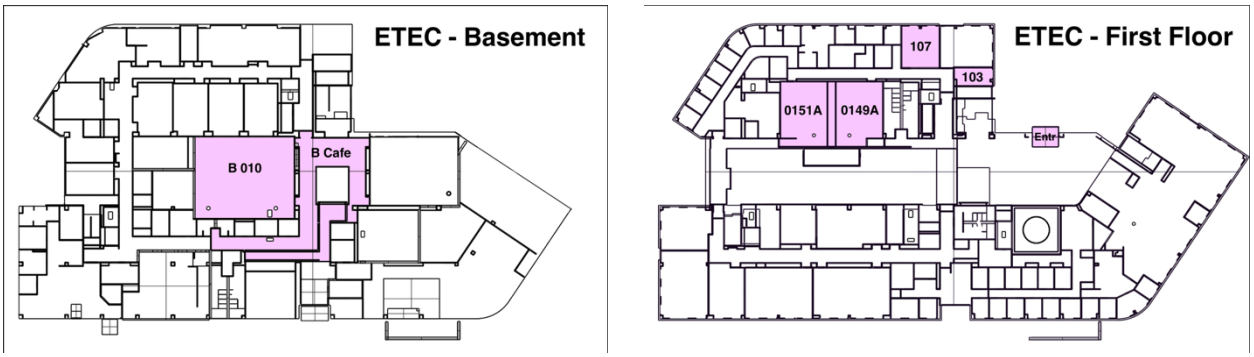


Fig 4. Top-down layout of basement floor (left) and first floor (right) of ETEC. Rooms part of the analysis is highlighted. Note that room 0151A and 0149A are connected as a large conference room.

In addition to $PM_{2.5}$, the sensors measured parameters including temperature, relative humidity, CO2 levels, barometric pressure, motion, and light level. These additional variables provided information on $PM_{2.5}$ level differences between daytime/nighttime and change in human activity.

The raw minute-level sensor data were processed and cleaned using the pandas library in Python and then aggregated into hourly averages. To investigate differences in indoor $PM_{2.5}$ conditions associated with building use and occupancy, two one-week periods in March 2026 were selected for detailed comparison: March 14–20, representing the academic break period, and March 22–28, representing the regular academic period.

For simulations, we used a 3D model of the ETEC building and airflow simulations were made on the Rhinoceros version 8 (Rhino 8) software, along with Grasshopper, Rhino's visual programming environment. Grasshopper plug-in contam-in-ANT (ANT) version 0.2.7 was used to establish the multizone airflow and contaminant transport model. ANT integrates the capabilities of National Institute of Standards and Technology (NIST)'s CONTAM simulation engine with Rhino/Grasshopper, allowing geometric information from the building model to be

translated into simulation inputs, including individual thermal/airflow zones, zone volumes, surface areas, and airflow connections. In addition, external factors such as human population, respiration, and viral spread, can be modeled as well. Within the ETEC model, five indoor spaces and one outdoor space were defined as separate zones so that variations in $PM_{2.5}$ concentrations among rooms could be evaluated.

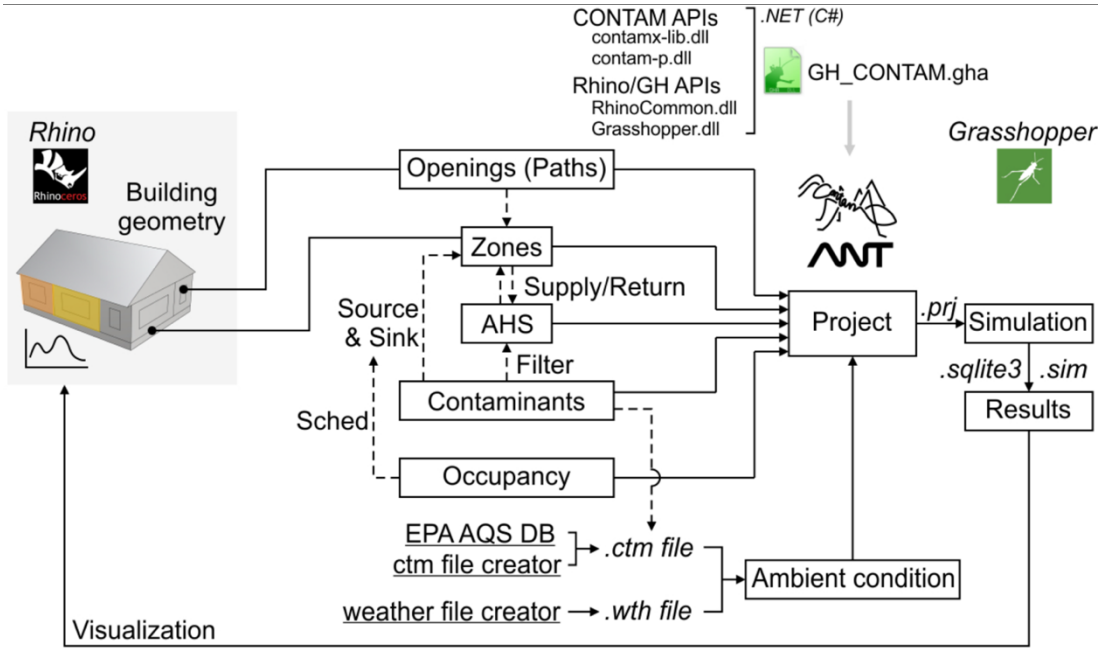


Fig 5. Example workflow for the ANT Grasshopper Plugin

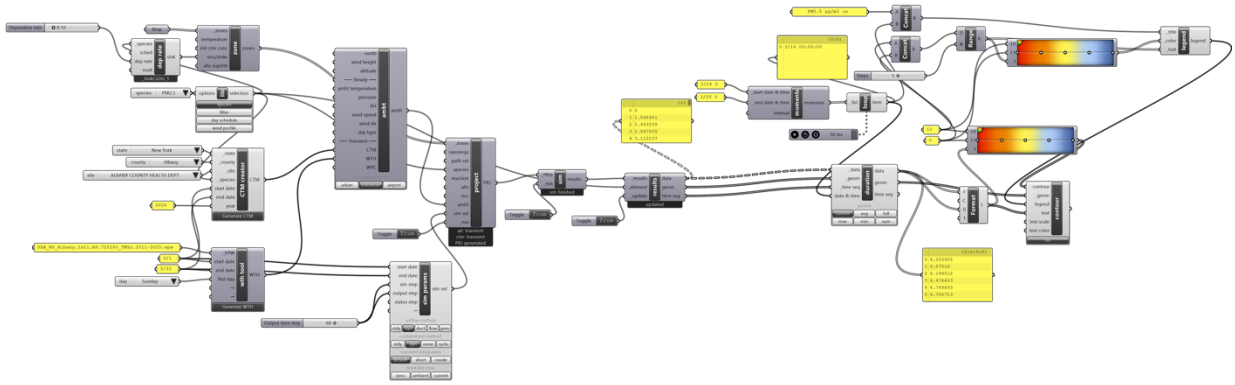


Fig 6. Preliminary model process in ANT Grasshopper

The simulation workflow was then configured to model the infiltration and transport of outdoor $PM_{2.5}$ into the building. Outdoor $PM_{2.5}$ concentration data were incorporated through ANT using data obtained from the U.S. Environmental Protection Agency (EPA) Air Quality System (AQS). ANT converts data from EPA into contaminant files, which are used for simulations. The weather data was based on the 2026 March weather data from the Open-Meteo API, manually created into a TMY file. The weather data provide time-varying meteorological parameters, including outdoor temperature, atmospheric pressure, wind speed, and wind direction. All these inputs are combined through the plug-in to perform transient multizone simulations.
Finally, simulation results were visualized as an animation directly on the ETEC building model within Rhino 8 using ANT's visualization components.

Minute-level indoor air quality data were collected at five indoor locations and one outdoor location within the ETEC building at the University at Albany. Indoor measurements were obtained using Milesight AM319 Ambient Monitoring Sensors, with two sensors located in small classrooms/conference rooms, two in large conference rooms, and one in a large café area that also includes a kitchen and hallway. Outdoor conditions were monitored using a Milesight AM308 Ambient Monitoring Sensor at the building entrance.

In addition to $PM_{2.5}$, the sensors measured environmental parameters including temperature, relative humidity, PM_{10}, CO_2, barometric pressure, motion, and light level. These additional variables provided information $PM_{2.5}$ level differences between daytime/nighttime differences and human activity. The raw minute-level sensor data were processed and cleaned using the pandas library in Python, and then aggregated into hourly, weekly, and monthly averages. To investigate differences in indoor $PM_{2.5}$ conditions associated with building use and occupancy, two one-week periods in March 2026 were selected for detailed comparison: March 14–20, representing the academic break period, and March 22–28, representing the regular academic period.

3. Preliminary Results and Discussion

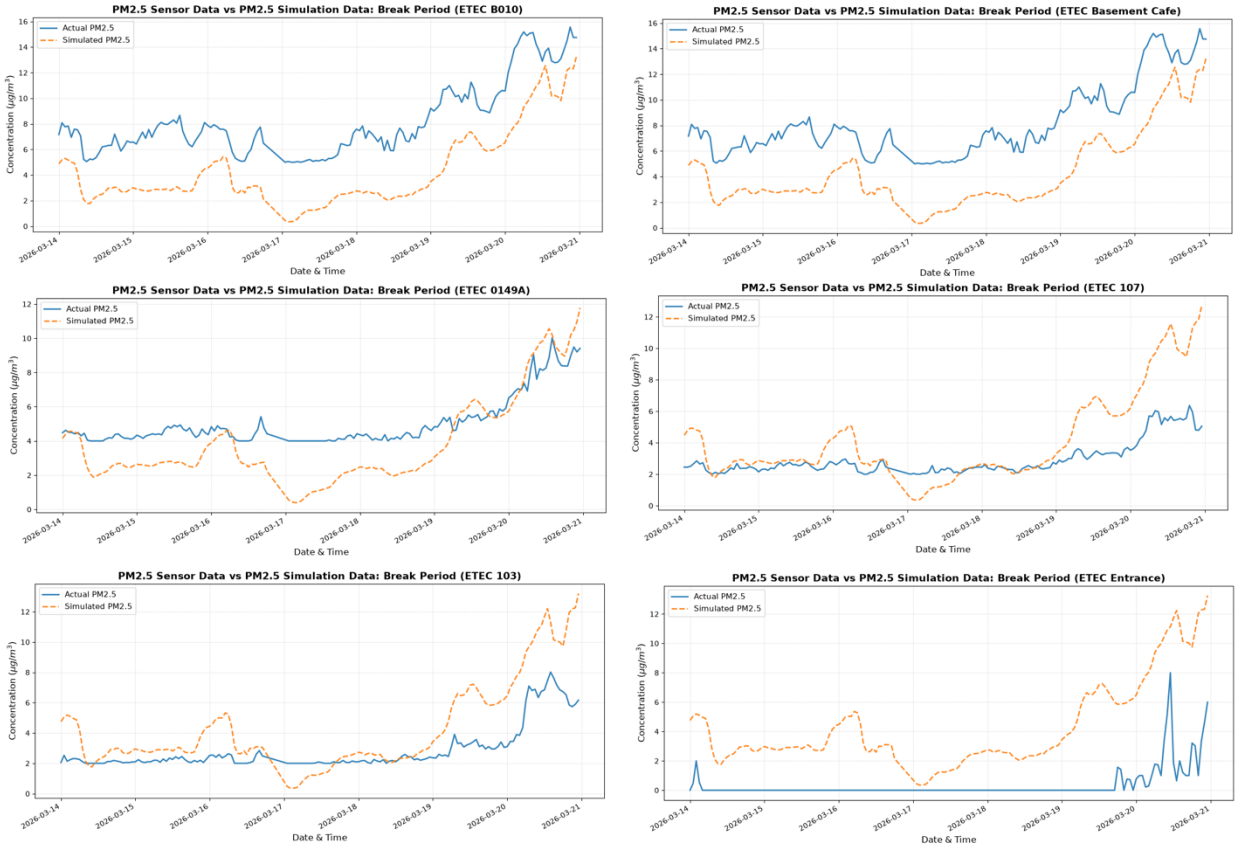


Fig 7. Time series of $PM_{2.5}$ data for a week of break period (March 14 to March 20) at ETEC. Actual $PM_{2.5}$ refers to $PM_{2.5}$ levels captured by sensors. Simulated $PM_{2.5}$ refers to $PM_{2.5}$ levels simulated through ANT.

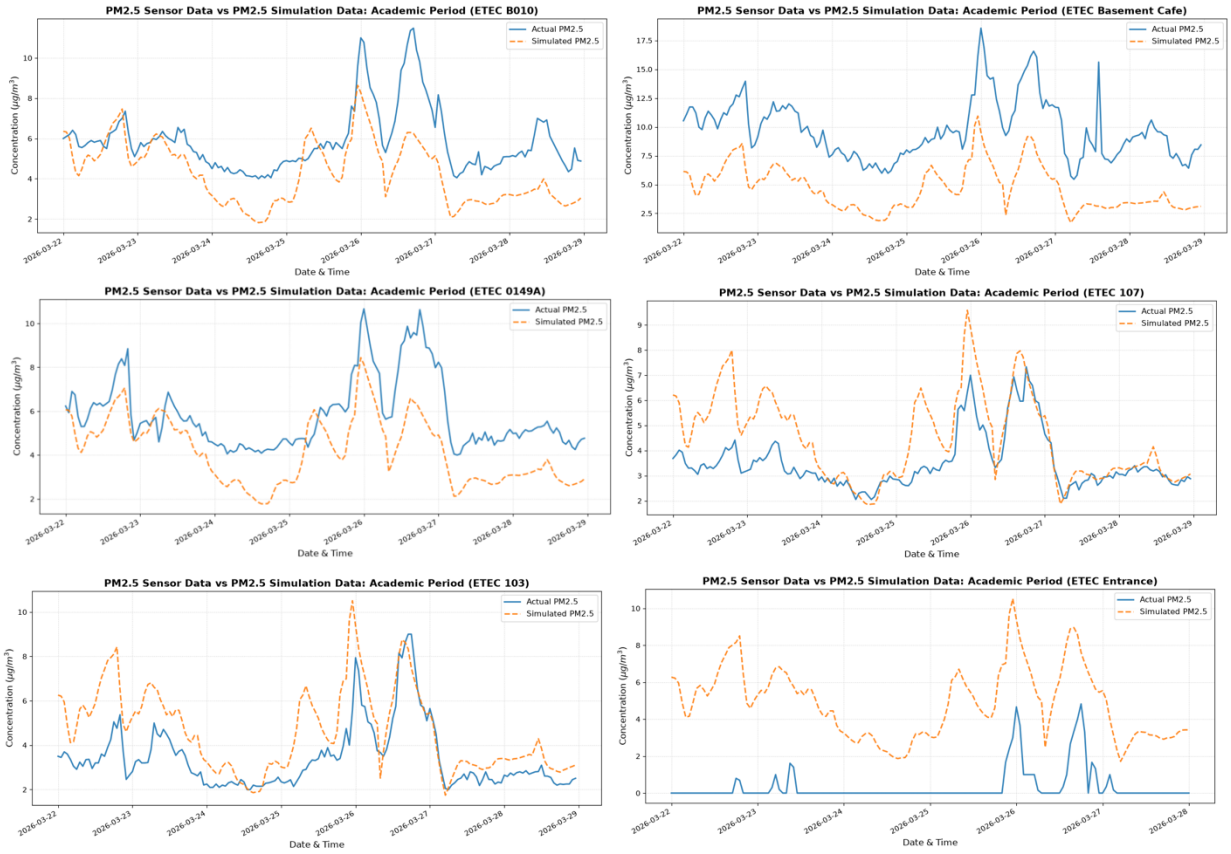


Fig 8. Time series of $PM_{2.5}$ data for a week of academic period (March 22 to March 28) at ETEC. Actual $PM_{2.5}$ refers to $PM_{2.5}$ levels captured by sensors. Simulated $PM_{2.5}$ refers to $PM_{2.5}$ levels simulated through ANT.

Overall, indoor $PM_{2.5}$ concentrations were higher during the regular academic period than during spring break, with similar seasonal patterns observed in both the sensor measurements and simulation results. Based on the sensor data, the mean indoor $PM_{2.5}$ concentration was 5.70 μg/m³ during the academic period, compared with 4.72 μg/m³ during spring break, representing an approximately 20.9% higher concentration during the academic period.

During the academic period, weekday and weekend concentrations were relatively similar, averaging 5.76 and 5.56 μg/m³, respectively. In contrast, a larger weekday–weekend difference was observed during spring break, with weekday and weekend concentrations averaging 5.00 and 4.05 μg/m³, respectively.

The simulation results showed lower overall indoor $PM_{2.5}$ concentrations than the sensor measurements during both periods, with mean concentrations of 4.52 μg/m³ during the academic period and 3.87 μg/m³ during spring break, compared with 5.70 μg/m³ and 4.72 μg/m³ from the sensor measurements, respectively.

During the academic period, the simulated weekday and weekend concentrations were 4.49 μg/m³ and 4.60 μg/m³, respectively, showing very little difference between the two periods and a slightly higher concentration on weekends. This pattern differs from the sensor measurements, with concentrations for sensor data showing overall higher $PM_{2.5}$ data, 33.9 percent difference for academic period and 35.8 percent error for break period.

The differences between the measured and simulated weekday–weekend patterns may be partly attributed to the current model configuration, which does not account for the actual number and temporal variation of occupants and human activities within the building. In particular, the kitchen area, which has cooking activity that is not considered for the simulation, showed the highest simulation error with 51.6 percent error during the academic period and 49.5 percent error during break period.

In addition, the smaller weekday–weekend difference during the regular academic period may reflect more consistent building use throughout the week. The sensor measurements were consistently higher than the simulated concentrations, with the model underestimating the indoor mean by approximately 21% during the academic period and 18% during spring break. This difference indicates that outdoor $PM_{2.5}$ infiltration alone may not fully account for the measured indoor concentrations. Additional factors, including indoor sources, occupant activity, particle resuspension, and differences between modeled and actual ventilation conditions, may contribute to the measured concentrations. Although the sensors collected additional measurements of CO_2, motion, and light levels, these variables were not incorporated into the current simulation to represent human activity. These measurements could be integrated into the model to better characterize temporal variations in occupancy and human activity. Incorporating activity-related data may improve the representation of indoor particle generation and resuspension and help explain the differences between measured and simulated $PM_{2.5}$ concentrations.

For future research, the original 3D building model could be further developed to represent these six spaces as individual zones and incorporate more detailed building information. Additional details related to room geometry, such as building material, ventilation, HVAC operations, connections between rooms and hallways, door openings, occupancy, and indoor activities could be integrated into the model. The measured CO_2, motion, and light-level data could also be incorporated to better represent temporal variations in human activity. By combining detailed building information with sensor-based activity data, the improved 3D model could provide a more realistic representation of $PM_{2.5}$ transport, indoor particle generation, and outdoor infiltration across the different spaces.

Conclusion

This study investigated the spatial and temporal variation of indoor $PM_{2.5}$ concentrations within the University at Albany ETEC building by combining distributed sensor measurements with multizone airflow and contaminant-transport simulations. Indoor $PM_{2.5}$ concentrations were generally higher during the regular academic period than during spring break, while differences between weekdays and weekends were more pronounced during the break period. The room-to-room comparison also demonstrated substantial differences, with the basement café and kitchen exhibiting the highest sensor-measured $PM_{2.5}$ concentrations, showing the influence of indoor activities such as cooking. The comparison between measured and simulated concentrations further indicated that outdoor particle infiltration alone does not fully explain the observed indoor $PM_{2.5}$ levels. The current model provides an initial setup for understanding particle transport within the building, but additional building and human-activity information is needed to better represent real indoor conditions.

Future work will focus on developing more detailed building models and expanding the scope of the measurements and simulations. First, simulations will be conducted using more detailed representations of individual rooms, incorporating building openings, interior features, HVAC systems, door-opening and closing frequency, and building materials. These additions will allow the model to better represent airflow and particle transport within individual spaces. Second, additional human-activity information will be incorporated by using sensor-based motion data and CO2 data to estimate room population and occupancy patterns. This information can help

characterize particle generation and resuspension associated with human activity more accurately.

The study will also be extended to include sensor and simulation data from later periods of the year. Measurements during the Canadian wildfires in July 2026 will be examined to investigate potential anomalies in indoor-to-outdoor $PM_{2.5}$ ratios. Finally, the monitoring and modeling framework will be expanded beyond ETEC to other buildings at the University at Albany, including the Caskill and Massary buildings. Comparing multiple buildings will provide an opportunity to evaluate how differences in building design, ventilation, occupancy, and room characteristics influence indoor $PM_{2.5}$ and outdoor particle infiltration. We anticipate that these efforts will contribute to the development of a more comprehensive building-based framework for understanding and modeling indoor air quality.

Acknowledgements

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